DO NOW — RECALL

Lesson 117 · Waterfall and Agile Development

Name: _____ Class: _____ Date: _____

1. What is the difference between linker and loader?

2. What is the time complexity of sequential search in the worst case?

3. After pushing A then B onto a stack, which is popped first?

4. Why must a binary search operate on ordered data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Linker resolves modules/symbols; loader maps executable into memory to run.

2. Model answer

O(n).

3. Model answer

B, because a stack is LIFO.

4. Model answer

It discards half the search space using an ordering comparison.

By the end, you can…

• Sequential analysis, design, implementation, testing and maintenance.

• Iterative delivery, feedback and reprioritisation in short cycles.

• Stability, regulation, stakeholder access and change risk affect suitability.

Vocabulary — know these terms

waterfall — Stage-based plan-driven development lifecycle.

agile — Iterative and adaptive development principles.

iteration — Repeated short cycle yielding feedback/increment.

requirements — Functions/constraints a system must satisfy.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Agile iterations with prioritised backlog, frequent releases and real user feedback. A strictly sequential plan risks discovering changed needs late. Maintain security and test discipline.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Waterfall lifecycle

Sequential analysis, design, implementation, testing and maintenance.

Agile methods

Iterative delivery, feedback and reprioritisation in short cycles.

Choose on evidence

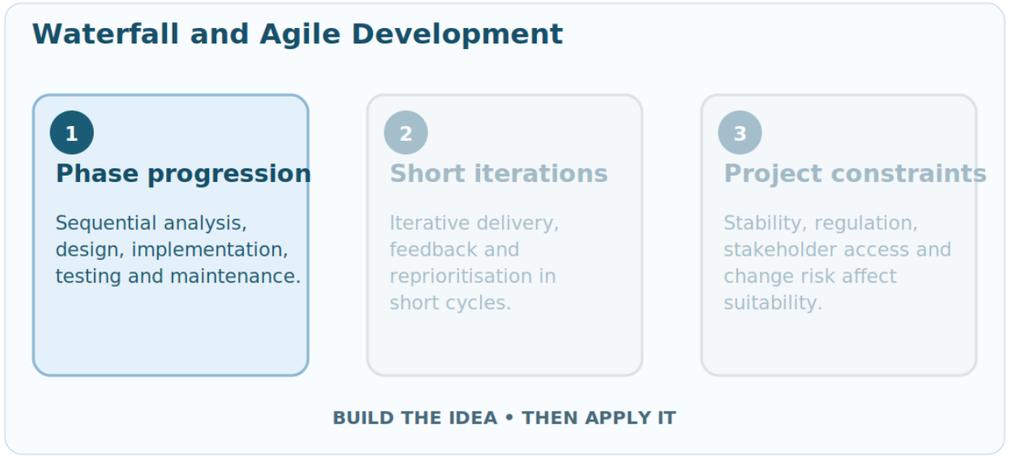
Stability, regulation, stakeholder access and change risk affect suitability.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

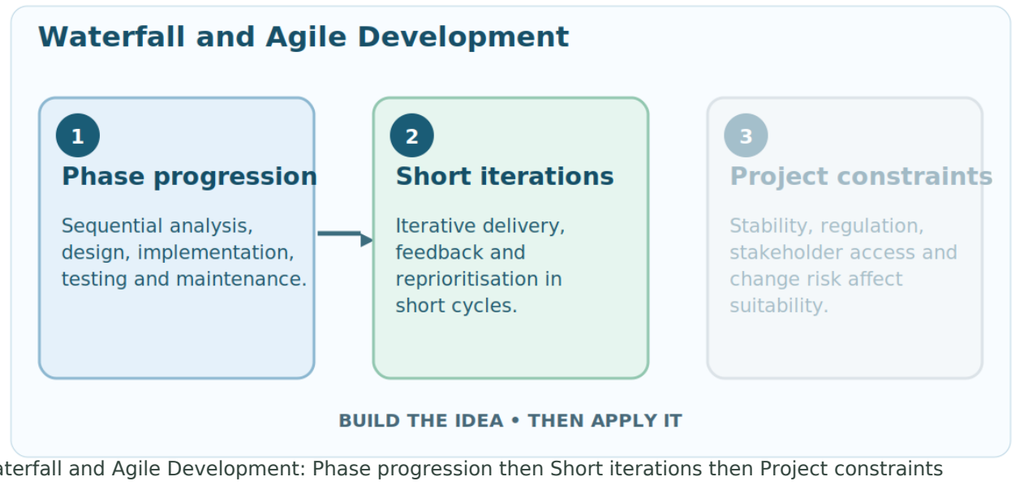
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Waterfall and Agile Development: Phase progression then Short iterations then Project constraints

New Learning



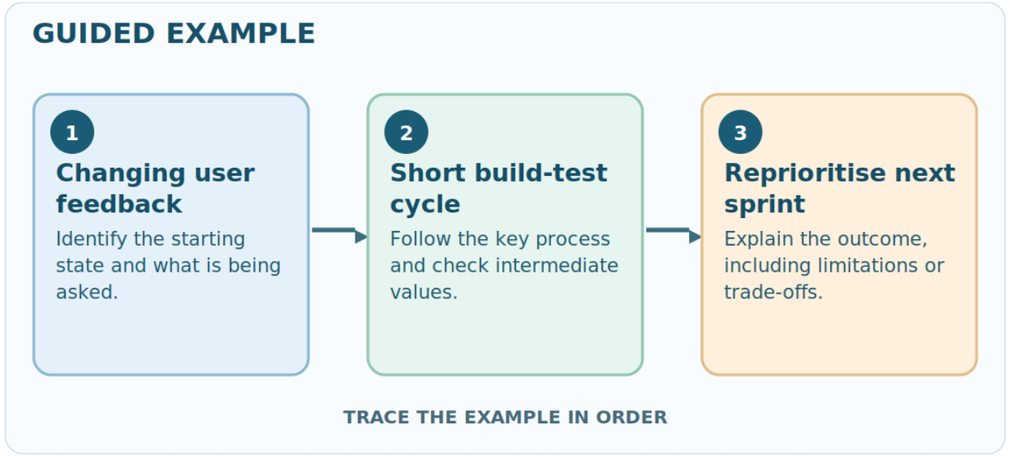
Waterfall and Agile Development: Phase progression then Short iterations then Project constraints

We Do Check · original lesson diagrams

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A community app changes weekly after users test prototypes. Which approach is more suitable?

Identify the inputs, relevant facts and exact output or judgement required.



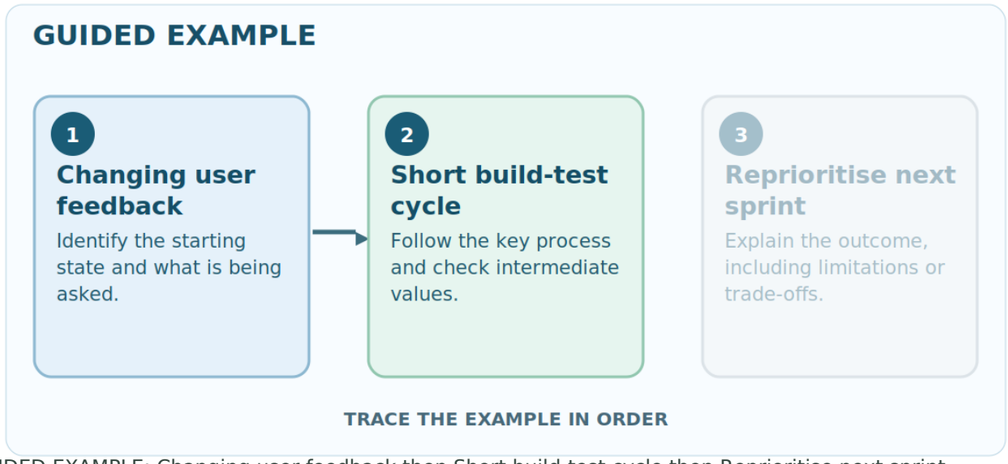
GUIDED EXAMPLE: Changing user feedback then Short build-test cycle then Reprioritise next sprint

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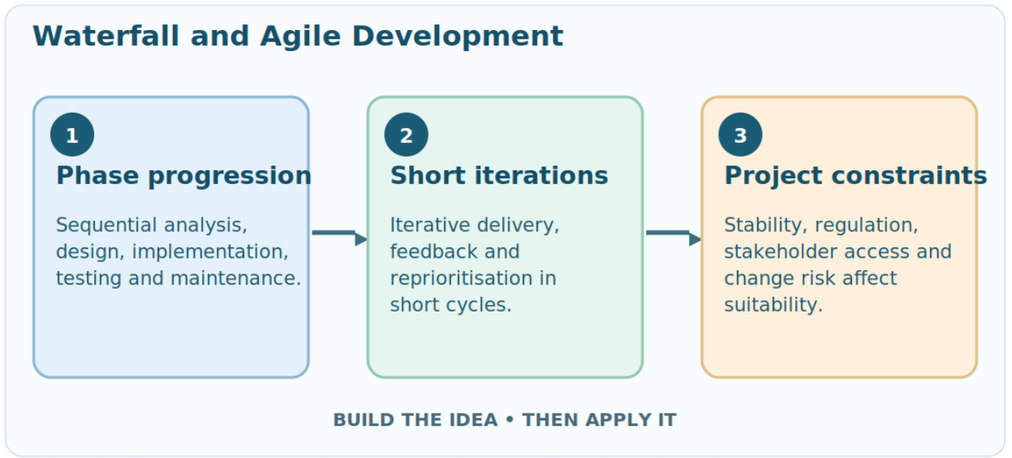


GUIDED EXAMPLE: Changing user feedback then Short build-test cycle then Reprioritise next sprint

New Learning · original lesson diagrams

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New Learning



Waterfall and Agile Development: Phase progression then Short iterations then Project constraints

I DO — WORKED EXAMPLE

Follow the worked method

An air-traffic subsystem has carefully fixed specifications and requires formal approvals. Select a development approach.

Model answer

A plan-driven/waterfall approach can support documented phase gates and traceability; however, testing and validation must still occur throughout as appropriate. Safety assurance matters more than simply labelling it waterfall.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Agile means a project needs no testing or documentation.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

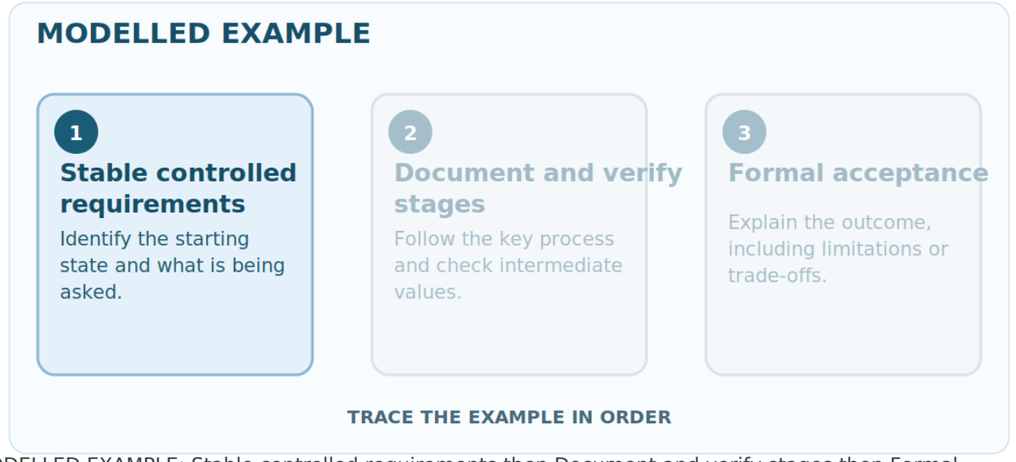
Evaluate waterfall versus agile for replacing an exam-management system subject to fixed deadlines and changing rules. [8]

I Do · original lesson diagrams

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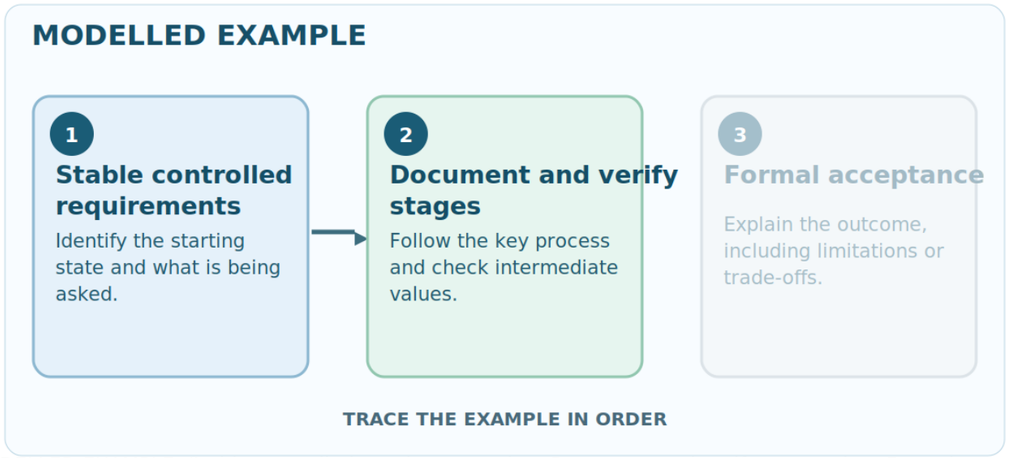
MODELLED EXAMPLE: Stable controlled requirements then Document and verify stages then Formal acceptance

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YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe the waterfall stages and one drawback of late requirement changes. [6]

Question 2

Explain two features of agile development in a public-services app. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

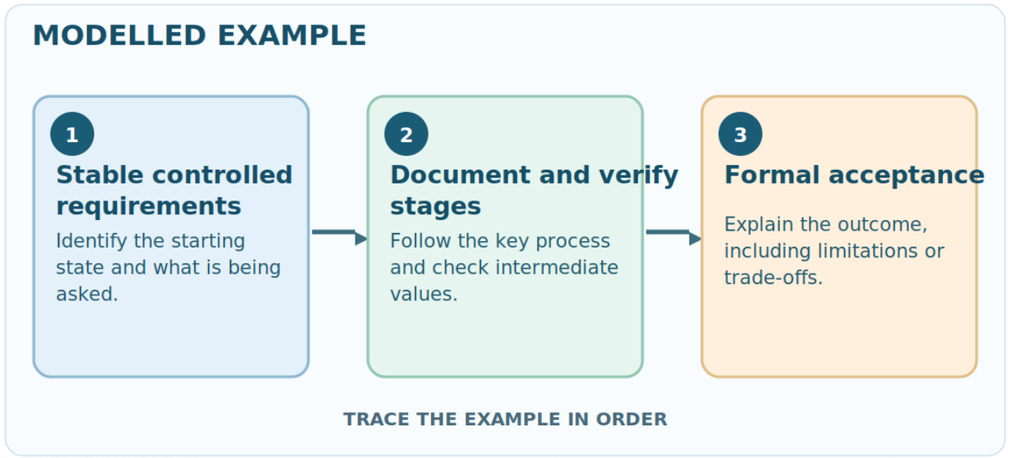
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

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